POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2019
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Consider an analog signal: 8

$$x_a(t) = 3\cos 100\pi t + 2\sin^2 180\pi t$$

i) What is Nyquist rate for this signal?

ii) Assume that the samples of a given signal is operated at 2100 bits/s and each input sample is quantized into 128 different voltage levels. What is the discrete-time signal obtained?

iii) Determine the time period of the discrete-time signal.

iv) What is the resolution Δ?

b) Illustrate the significance of convolution summation in digital signal analysis. Compute and plot the convolution of $x[n] = \{0, \underset{\uparrow}{1}, 5, 4, 0\}$ and $h[n] = \{5, 3, \underset{\uparrow}{2}, 1, 0\}$. 7

2. a) Resolve the discrete time signal into impulses. Derive the equation for convolution sum for discrete-time LTI system. 8

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b) Find the Z-transform of $x[n] = a^n \cos(w_0 n)u[n]$. Also specify ROC. 7

3. a) Find the 8-point DITFFT of $x(n) = \sin\frac{3\pi n}{4}, for\ 0 \le n \le 7$ 8

b) Show that the multiplication of DFTs of two sequences result in the circular convolution in time domain. 7

4. a) Obtain the parallel form realization of following IIR filter 7

$$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$

b) A LPF has the desired frequency response 8

$$H_d(e^{jw}) = \begin{cases} e^{-j3\omega} ; 0 \le |\omega| < \frac{\pi}{2} \\ 0 ; \frac{\pi}{2} < |\omega| \le \pi \end{cases}$$

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6 Determine the filter coefficients $h(n)$ for N=7 using type –I frequency sampling technique.

5, 6.3a) Show that the Kaiser window includes the rectangular window as a special case.

6.2b) Use Hanning window method to design low-pass FIR filter with pass-band edge frequency $(\omega_p) = 0.4\pi$ & M = 11. Where M is the filter length.

7.1 6. a) Design a digital lowpass Butterworth filter using Bilinear transformation method to meet the following specifications:

Passband attenuation 3 dB	Passband edge = 2 KHz
Stopband attenuation 30 dB	Stopband edge = 6 KHz

Use sampling frequency = 10 KHz.

7.2 b) Determine H(z) using impulse invariance method for the following analog filter's transfer function :

$$H_a(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

7. Write short notes on: **(Any two)**
   a) Energy Signal Vs Power Signal 3
   b) Computational Complexity of DFT 4
   c) Gibbs Phenomena 6.1

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